

Deliverable 1

What is a database?

A database is a collection of data that can be stored, managed, and accessed.

What is a Linux container?

A Linux container is a lightweight, isolated environment that runs applications using the host system's kernel, similar to a virtual machine but with less overhead.

What is Proxmox VE?

Proxmox VE is an open-source virtualization platform that lets you run virtual machines and containers on a server.

What is Debian?

Debian is a open source Linux operating system known for stability.

What is Mariadb?

Mariadb is a open source database management system used to store and manage structured data.

What is SQL?

SQL (Structured Query Language) is a programming language used to manage and query data in relational databases like Mariadb & MySQL.

What is a home lab?

A home lab is a personal setup where you experiment with servers, networking, virtualization, and IT tools to learn and practice skills.

How to create your own home lab with Proxmox VE

1) Pick the hardware

- PC, mini PC, or old server
- Ethernet connection (Its better if you can)
- Enough RAM and storage for VMs/containers
- Extra drive for backups(Optional)



HP EliteDesk 800 G3 Mini Desktop Computer, Intel Quad-Core i5-6500 Business PC, 16GB DDR4 RAM, 256GB SSD, Keyboard&Mouse, WIFI, BT, Windows 11 Pro (Renewed)

3.7 ★★★★★ (15)
100+ bought in past month

\$154⁰⁰ List Price: \$199.99

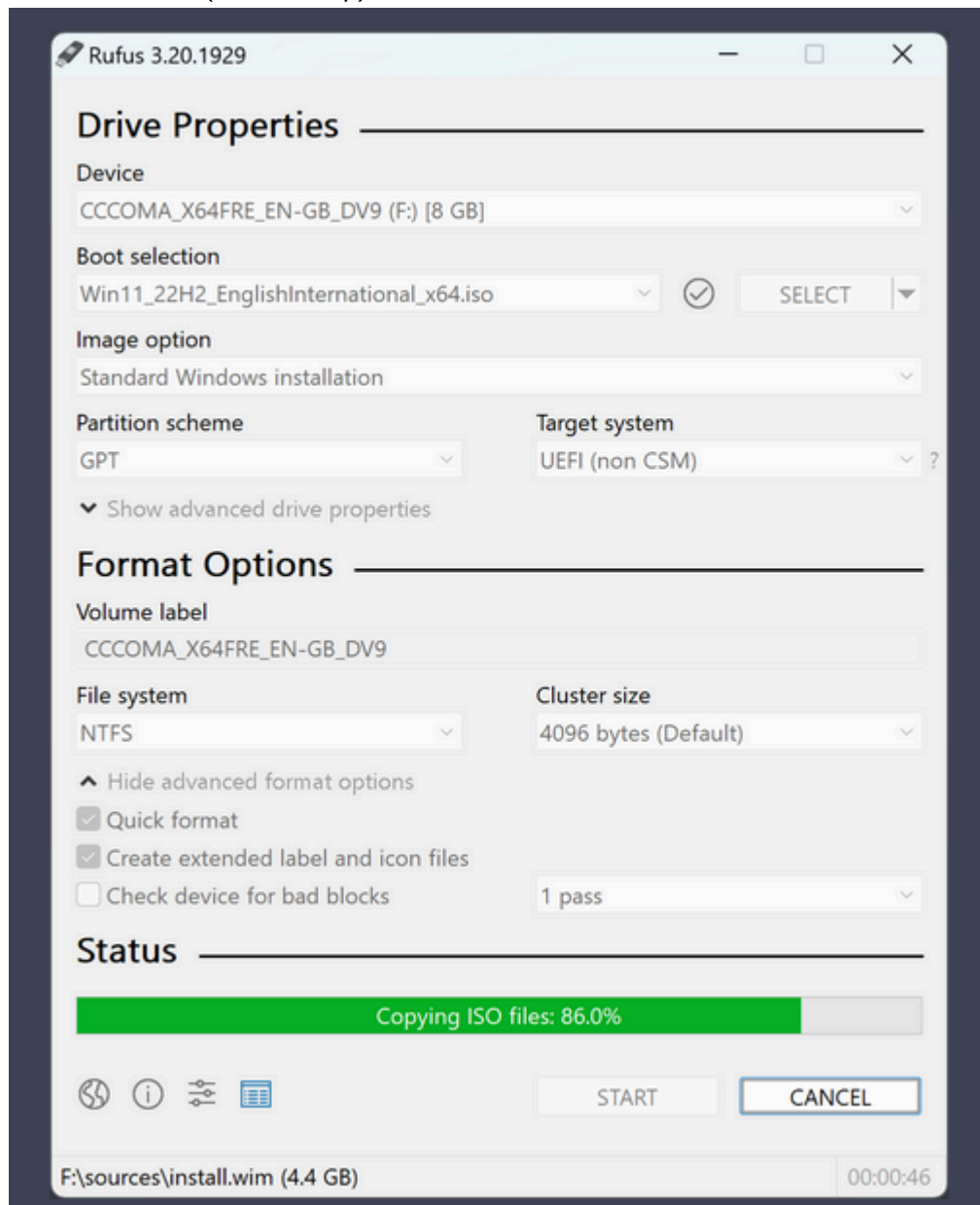
FREE delivery **Tue, Dec 23**
Arrives before Christmas
Only 2 left in stock - order soon.

See options

Processor Type	Disk Size	RAM	Processor Speed
Core i5	256 GB	-	3.2 GHz

2) Download required tools

- Proxmox VE ISO (<https://www.proxmox.com/en/downloads/proxmox-virtual-environment/iso>)
- Rufus (USB installer)
- USB flash drive (8GB and up)



3) Create the Proxmox USB with Rufus

1. Plug in the USB drive
2. Open Rufus
3. Device: select your USB

4. Boot selection: choose the Proxmox VE ISO
5. Click Start
6. Unplug the USB when finished

4) Boot from the USB

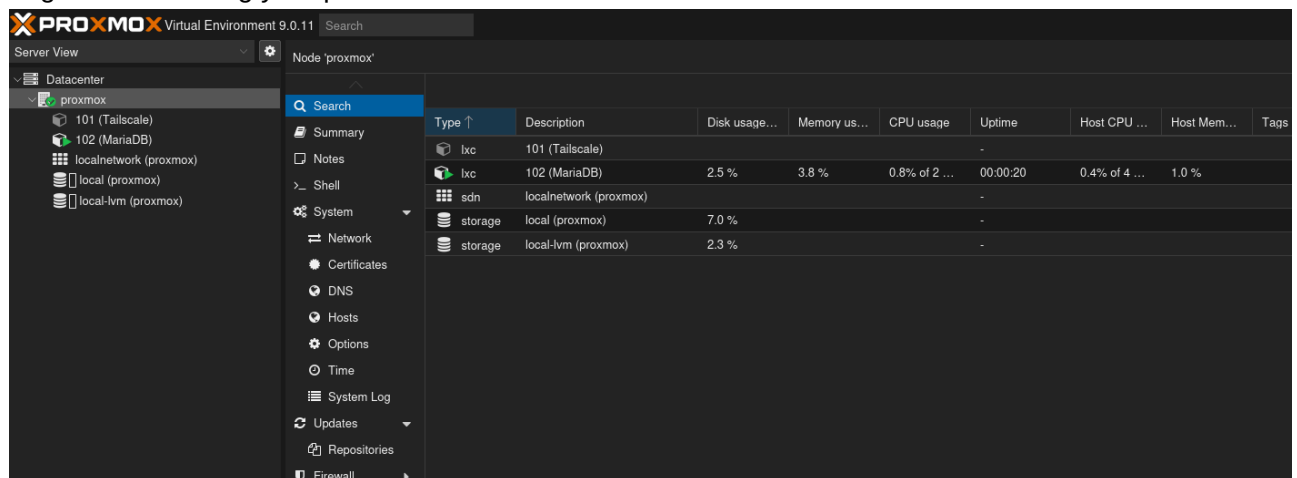
1. Insert the USB into the server
2. Power on and open the Boot Menu / BIOS
3. Select the USB drive

5) Install Proxmox VE

1. Select Install Proxmox VE
2. Accept the license
3. Choose the installation disk
4. Set country, keyboard, and timezone, etc
5. Create a password and enter an email
6. Configure networking
 - Set a static IP
 - Example: "192.168.1.50"
7. Complete installation and reboot
8. Remove the USB drive

6) Access the Proxmox web

1. open a browser (On another computer)
2. Go to https://PROXMOX_IP_HERE (Meaning the ip you gave it):8006
3. Log in as root using your password



4.

7) Basic Proxmox setup

- Update the system:
 - apt update
 - apt upgrade

8) Create your first VM/Container

- Use containers for lightweight services

- Use VMs for full operating systems Steps:
1. Download a Debian template (container) or ISO (VM)
 2. Create the VM/CT
 3. Assign CPU, RAM, and disk
 4. Start the VM/CT

The screenshot shows the 'Create: LXC Container' wizard in Proxmox VE, specifically the 'General' tab. The interface is dark-themed. At the top, there's a title bar with a close button. Below it, a series of tabs: 'General' (active), 'Template', 'Disks', 'CPU', 'Memory', 'Network', 'DNS', and 'Confirm'. The main area contains several input fields and checkboxes. On the left, 'Node:' is set to 'proxmox', 'CT ID:' is '100', 'Hostname:' is empty, 'Unprivileged container:' is checked, and 'Nesting:' is checked. On the right, 'Resource Pool:' is empty, 'Password:' is empty (highlighted with a red border), 'Confirm password:' is empty, and 'SSH public key(s):' is empty. A 'Load SSH Key File' button is below the SSH key field. At the bottom, there's a 'Help' button, an 'Advanced' checkbox, and 'Back' and 'Next' buttons.

5.

10) Install services (MariaDB in this case)

- Install MariaDB on Debian
- Create databases and users

11) Extra step: Adding tailscale (Do access Server outside your network)

- Install tailscale in the server
- run the command: 'tailscale up'
- use the link given to add server to tailnet
- Add the device you want to use to connect to the tailnet

MACHINE
alexpc-1 xander.castro1@gmail.com
cis106 xander.castro1@gmail.com
mariadb xander.castro1@gmail.com
proxmox xander.castro1@gmail.com
tailscale xander.castro1@gmail.com
SSH

- connect using the IP address give by tailscale

12) Other home lab projects:

- MariaDB server
- Nextcloud personal cloud
- Email server
- Web server
- Windows VM